

CENG 222

Statistical Methods for Computer Engineering

Spring 2016-2017

Section 1

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Section 1 Course Web Page:

http://www.ceng.metu.edu.tr/~tcan/ceng222_s1617

Goals of the course

- Learn techniques and tools to be able to:
 - analyze and interpret large scale data,
 - apply probability theory and statistics to handle uncertainty,
 - infer facts and relationships from collected data, and
 - construct simulations by sampling from arbitrary distributions
- Acquire skills for the hot new CS field: “Data Science”

Course outline

- See the tentative schedule at:
 - http://user.ceng.metu.edu.tr/~tcan/ceng222_s1617/Schedule/index.shtml

Grading

- Midterm exam - 40%
- Final exam - 40%
- 4 Assignments (5% each) - 20%

Section 1 Course Web Site

- Syllabus
- Lecture slides and reading materials

COW

- Assignments
- Announcements at the news group: `course.222`
- We may also use ODTU-Class for announcements and assignments

Textbook

- Probability and Statistics for Computer Scientists, Second Edition, Michael Baron, 2013
- Your main resource of study for this course

Probability

- Studies uncertainty
- A random experiment
 - An experiment/observation which does not have a certain outcome before it is conducted
 - Examples
 - Tossing a coin
 - Observing the life time of a light bulb
 - Number of games the Cavaliers will win this season
 - Others?

Sample space

- The set of all possible outcomes of a random experiment is called the sample space
 - Tossing a coin:
 - Sample space = {H, T}
 - Tossing two coins:
 - Sample space = {HH, HT, TH, TT}
 - Lifetime of a light bulb:
 - Sample space = $[0, +\infty)$

Event

- Any collection of possible outcomes of an experiment
 - Any subset of the sample space
- Examples:
 - Experiment: tossing two coins. Event: obtaining exactly one head. $\{HT, TH\} \subset \{HH, HT, TH, TT\}$
 - Experiment: lifetime of light bulb. Event: light bulb does not last more than a month.
 $[0, 1] \subset [0, +\infty)$

Event

- A sample space of N possible outcomes yields 2^N possible events
- Example: tossing a dice once
- Sample space = $\{1, 2, 3, 4, 5, 6\}$
- Number of possible events = $2^6 = 64$
- Example events?

Notation used in the book

- Ω = sample space
- $\emptyset$ = empty event
- $P\{E\}$ = probability of event E

Event algebra

- Union of two events: same as set union
 - $A \cup B = \{x: x \in A \text{ or } x \in B\}$
- Intersection of two events: same as set intersection
 - $A \cap B = \{x: x \in A \text{ and } x \in B\}$
- Complementation: same as in sets
 - A^c or $\bar{A} = \{x: x \in \Omega \text{ and } x \notin A\}$
- Difference: same as in sets
 - $A \setminus B = \{x: x \in A \text{ and } x \notin B\}$

Disjoint and exhaustive events

- Disjoint events: If A and B have no outcomes in common, i.e., $A \cap B = \emptyset$
 - Also called mutually exclusive events
- If the union of a number of events equals the sample space, they are called exhaustive
 - $A \cup B \cup C = \Omega$

Complement, Union, Intersection

- $\overline{A \cup B} = \bar{A} \cap \bar{B}$
- $\overline{A \cap B} = \bar{A} \cup \bar{B}$
- $\overline{E_1 \cup E_2 \cup E_3 \cup E_4} = \bar{E}_1 \cap \bar{E}_2 \cap \bar{E}_3 \cap \bar{E}_4$
- $\overline{E_1 \cap E_2 \cap E_3 \cap E_4} = \bar{E}_1 \cup \bar{E}_2 \cup \bar{E}_3 \cup \bar{E}_4$

Probability

- Assignment of a real number to an event
 - The relative frequency of occurrence of an event in a large number of experiments
- $P(A)$
- Axioms of probability:
 - $P(A) \geq 0$
 - $P(\Omega) = 1$
 - If A and B are mutually exclusive events, then
$$P(A \cup B) = P(A) + P(B)$$
- Any function that satisfies these axioms is called a probability function

Example

- Experiment:
 - Tossing two coins
 - $A = \{\text{obtaining exactly one head}\}$
 - $P(A) = ?$

Computing probabilities

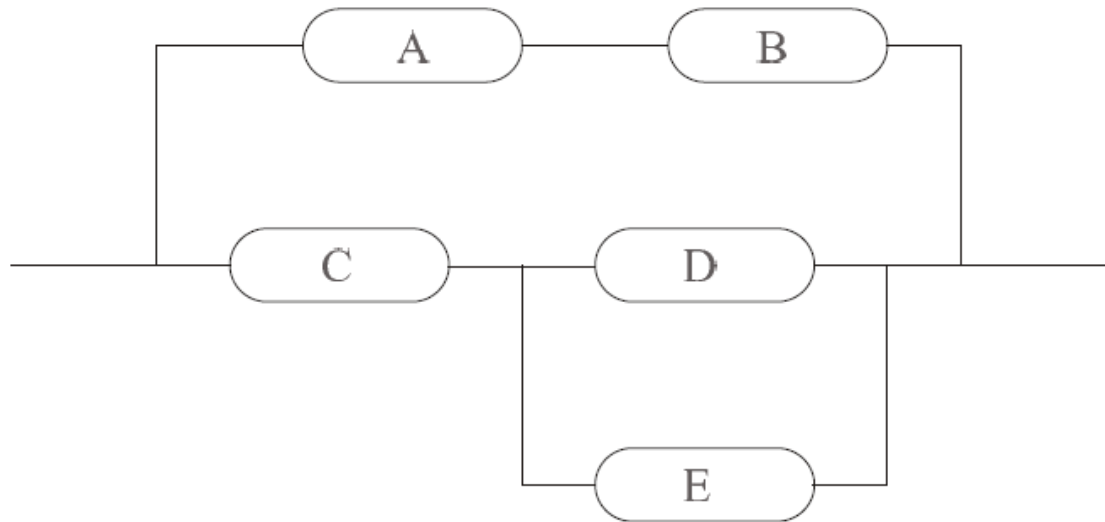
- for non-“mutually exclusive” events:
 - $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Independent Events

- $P(E_1 \cap E_2 \cap E_3) = P\{E_1\} \cdot P\{E_2\} \cdot P\{E_3\}$

Applications in reliability

- Example 2.18
- Example 2.19
- Example 2.20



Conditional probability

- Updating of the sample space based on new information
- Consider two events A and B . Suppose that the event B has occurred. This information will change the probability of event A .
- $P(A|B)$ denotes the conditional probability of event A given that B has occurred.

Conditional probability

- If A and B are events in Ω and $P(B) > 0$, then $P(A|B)$ is called the conditional probability of A given B if the following axiom is satisfied:
 - $P(A|B) = P(A \cap B) / P(B)$
- Example: tossing a fair dice.
 - $A = \{\text{the number on the dice is even}\}$
 - $B = \{\text{the number on the dice} < 4\}$
 - $P(A|B) = ?$

Independence

- If $P(A|B)=P(A)$ we call that event A is independent of event B
- Note:
 - if two events A and B are independent, then
$$P(A \cap B) = P(A)P(B)$$
- Show that $P(B|A)=P(B)$ also holds in this case.
 - In other words, A and B are mutually independent
- This does NOT mean that they are disjoint. If A and B are disjoint then $P(B|A)=0$

Independence

- Example: tossing a fair dice.
 - $A = \{\text{the number on the dice is even}\}$
 - $B = \{\text{the number on the dice} > 2\}$
 - $P(A|B) = ?$
 - $P(B|A) = ?$
 - $P(A) = ?$
 - $P(B) = ?$
- Example 2.31

Bayes' Rule

- Using conditional probability formula we may write:
 - $P(A|B) = P(A \cap B) / P(B)$
 - $P(B|A) = P(A \cap B) / P(A)$
 - $\rightarrow P(A \cap B) = P(A|B)P(B) = P(B|A)P(A) \rightarrow$
 $P(B|A) = P(A|B)P(B) / P(A)$
- This is known as the Bayes' rule
- It forms the basis of Bayesian statistics
- What additional probabilities do we need to know to solve Example 2.32?

Law of Total Probability

- Let $B_1, B_2, B_3, \dots, B_k$ be a partition of the sample space. B_i s are mutually disjoint. Let A be any event.
- Note that B_i s also partition A
- Then for each $i = 1, 2, \dots, k$

$$P(B_i | A) = \frac{P(A | B_i)P(B_i)}{P(A)} = \frac{P(A | B_i)P(B_i)}{\sum_{j=1}^k P(A | B_j)P(B_j)}$$

When $P(A)$ is not directly known, but known conditionally, we make use of this law.

Bayes' Rule for two events

$$P(B \mid A) = \frac{P(A \mid B)P(B)}{P(A)} = \frac{P(A \mid B)P(B)}{P(A \mid B)P(B) + P(A \mid \bar{B})P(\bar{B})}$$

- Now, solve Exercise 2.32, given $P(B)$

Another example

- A novel disease diagnostic kit is 95% effective in detecting a certain disease when it is present. The test also has a 1% false positive rate. If 0.5% of the population has the disease, what is the probability a person with a positive test result actually has the disease?

Solution

- $A = \{\text{a person's test result is positive}\}$
- $B = \{\text{a person has the disease}\}$
- $P(B) = 0.005, P(A|B) = 0.95, P(A|B^c) = 0.01$

$$P(B|A) = \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|B^c)P(B^c)}$$
$$= \frac{0.95 \times 0.005}{0.95 \times 0.005 + 0.01 \times (1 - 0.005)} = \frac{475}{1470} \approx 0.323$$

Random Variables

- A random variable (r.v.) associates a unique numerical value with each outcome in the sample space. It is a real-valued function from a sample space Ω into real numbers.
- Similar to events it is denoted by an uppercase letter (e.g., X or Y) and a particular value taken by a r.v. is denoted by the corresponding lowercase letter (e.g., x or y).

Examples

- Toss three coins. X = number of heads
- Pick a student from the Computer Engineering Department.
 X = age of the student
- Observe lifetime of a light bulb
 X = lifetime in minutes
- X may be discrete or continuous